

Mudit Jain

Senior Deep Learning Engineer | E2E/VLA architectures, AV Perception, and Planning

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Professional Summary

Senior Deep Learning Engineer with 8+ years of experience building real-time autonomous-driving perception and planning systems. At Qualcomm, currently working on end-to-end VLA/VA architectures for autonomous driving, including map-conditioned multimodal context fusion for planning; previously delivered BEV camera+LiDAR perception models across diverse global markets with 90% auto-annotation yield for 2D/3D detection.

Core Skills

Programming: C++ (8+ years), Python (8+ years)

ML/DL Frameworks: PyTorch, Transformers, PEFT, HuggingFace, Ray

Perception & Autonomy: VLA/VA architectures, 2D/3D object detection and tracking (LiDAR + Camera), sensor fusion, BEV world modeling, map-conditioned planning, end-to-end AV architectures

Systems: Real-time perception, embedded/automotive platforms, ISO 26262, SLAM, non-linear optimization, model and data parallelism

Experience

Qualcomm, San Diego

2021 – Present

Senior Deep Learning Engineer – AV Perception / Planning / VLA

- Designed and implemented a decoder for a VLA architecture, including multimodal token fusion, architecture iteration, scaling-law exploration, data discretization, loss-function ablations, training-schedule optimization, and rigorous evaluation across diverse driving scenarios; improved trajectory prediction quality by ~30% on ADE/FDE metrics while enabling stronger long-horizon planning consistency and implicit scene understanding.
- Scaled autonomous-driving VLA model training across 100+ GPUs, sustaining 90%+ GPU utilization by optimizing large-scale distributed training systems for massive driving datasets, including data-loader throughput, model execution backends, distributed data parallelism, GPU memory utilization, performance monitoring, stall diagnostics, and training workflow stability.
- Mentored and technically guided Qualcomm Innovation Fund research projects, including an end-to-end autonomous-driving architecture project accepted at ECCV 2026 and a 3D object detection project accepted at the CVPR 2026 Workshop.
- Developed and fine-tuned BEV-based multimodal perception models using a 5-camera + 1-LiDAR sensor setup across diverse driving domains in India, UAE, Europe, and the U.S.; achieved ~90% yield in automated 2D/3D object-detection annotation workflows, exceeding internal performance targets.

Google Summer of Code (OpenCV)

2024 – Present

Developer & Mentor – 3D Reconstruction

- Developed a pipeline converting unconstrained video into efficient Gaussian splats; collaborated with OpenCV under Gary Bradski's mentorship.

Drone Lab, UC San Diego

2019 – 2021

Graduate Student Researcher (Funded)

- Designed and deployed an attention-based CNN using data from 600 cameras for wildfire plume detection in the ALERTWildfire initiative.

NVIDIA, Bengaluru

2016 – 2019

Embedded System Software Engineer

- Led design, implementation, and testing of an I2C virtualization software pipeline on embedded automotive platforms aligned with ISO 26262 functional safety requirements.

Education

University of California San Diego — M.S. in ECE (Machine Learning & Data Science)

BITS Pilani — B.E. in ECE

Publications, Awards, and Service

Papers:

BeyondSight: Object Permanence for End-to-End Autonomous Driving (*ECCV 2026*)

FishRoPE: Adapting Vision Foundation Models to Fisheye Geometry (*OmniCV, CVPR 2026*)

Additional:

6 patents submitted

2 Qualcomm Distinguished Awards

Reviewer: ECCV, WACV, IEEE IV, ITSC

GSoC Mentor (3D reconstruction)

Judge/Mentor, Qualcomm Innovation Fund